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BRIEF REPORT



Cognitive flexibility and resilience measured through a residual approach

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ABSTRACT

Background and Objectives: Resilience refers to the process through which individuals show better outcomes than what would be expected based on the adversity they experienced. Several theories have proposed that variation in resilience is underpinned by cognitive flexibility, however, no study has investigated this using an outcome-based measure of resilience.

Design: We used a residual-based approach to index resilience, which regresses a measure of mental health difficulties onto a measure of adversity experienced. The residuals obtained from this regression constitute how much better or worse someone is functioning relative to what is predicted by the adversity they have experienced.

Methods: A total of 463 undergraduate participants completed questionnaires of mental health difficulties and adversity, as well as a number-letter task-switching task to assess cognitive flexibility.

Results: Multiple regression analyses showed that better cognitive flexibility was not associated with greater resilience.

Conclusions: Our findings do not support theoretical models that propose the existence of a relationship between cognitive flexibility and resilience. Future research may serve to refine the residual-based approach to measure resilience, as well as investigate the contribution of “hot” rather than “cold” cognitive flexibility to individual differences in resilience.

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Introduction

The lifetime prevalence of traumatic events and other life events that can have a negative psychosocial impact is over 70% (Knipscheer et al., 2020). While exposure to such negative life events can profoundly affect behavioral and emotional functioning, there exists a large variation in the degree to which individuals are affected by adverse experiences (Bonanno, 2004). This is referred to as resilience. Specifically, resilience can be explained as “the phenomenon that many people do not or only temporarily become mentally ill despite significant psychological or physical burden” (Chmitorz et al., 2018, p. 78).

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While historically resilience was seen as a stable and intrinsic attribute of a person (i.e., a trait), more recent outcome-oriented approaches to resilience define resilience as the extent to which individuals can continue to function adaptively in the face of adversity (Chmitorz et al., 2018). Critically, resilience can be differentiated from emotional vulnerability in that in order to show resilience an individual has to have experienced adversity, and their emotional functioning must be evaluated in light of that adversity (Kalisch et al., 2021). As such, resilience can be operationalized as the variation in the degree to which individuals experience less emotional problems than what would be expected based on the adversity they experienced (Rutter, 2006). High resilience is associated with a range of positive outcomes, including enhanced physical health, and greater life satisfaction (Bonanno, 2004; Kong et al., 2015; Southwick et al., 2014). Consequently, it is important to enhance our understanding of the factors that contribute to variation in resilience, because such knowledge may facilitate our capacity to increase resilience in at-risk populations.

Various theories have proposed that cognitive flexibility could play a key role in such variation in resilience (Kashdan & Rottenberg, 2010; Parsons et al., 2016; Schwager & Rothermund, 2014). Cognitive flexibility refers to the executive function enabling individuals to shift between mental sets in response to changing environmental demands (Miyake et al., 2000). To illustrate how cognitive flexibility might promote resilience, imagine an emergency responder who is able to rapidly change their focus between tasks, for example coordinating the preservation of a scene and disseminating information to other services. Such flexibility may assist in feelings of self-efficacy, positive appraisal or reappraisal of the situation, and ultimately more resilient outcomes when faced with adversity.

In an influential paper, Kashdan and Rottenberg (2010) proposed that psychological flexibility is a fundamental determinant of health. Their definition of psychological flexibility includes the ability to adapt to changing situational demands and shift mindsets, and therefore overlaps with prior definitions of cognitive flexibility. The authors comprehensively reviewed evidence showing that psychological flexibility is associated with improved health and reduced psychopathology, and argued that this flexibility allows people to extract the best possible outcomes in a variety of situations. As such, this model is frequently cited as one of the first to emphasize the potential role of flexibility in the variation in resilience. Similarly, Schwager and Rothermund (2014) proposed that flexibility is a key mechanism underlying resilience. In their model, flexibility originates through a counter-regulation mechanism that leads cognitions to be biased toward information opposite in valence to the current context. Such flexible shifting to processing positive information in negative contexts and to negative information in positive contexts prevents rigidity or extremity in emotional functioning, and is thus suggested to promote resilience (Koole et al., 2015; Schwager & Rothermund, 2014).

These ideas are further echoed in a recent cognitive model of resilience. Parsons et al. (2016) proposed that flexibility in affective–cognitive systems is a key element to achieving resilient outcomes. In this model, a mapping system integrates information about the current situation and evaluates whether current responses are appropriate in light of individuals' goals. If the system's output is not conducive to reach desired goals, for example because the environment has changed, then the mapping system will switch to a different cognitive processing mode to promote a more adaptive outcome. By applying either persistence or flexibility depending on the environmental cues, outcomes and goals, individuals can maximize their ability to achieve the best possible outcomes.

Despite the theoretical propositions of a relationship between cognitive flexibility and resilience, limited empirical research supports the existence of such a relationship. For example, Genet and Siemer (2011) measured resilience as a stable personality trait using the self-reported ego-resilience scale (ER89; Block & Kremen, 1996) and the Connor-Davidson Resilience Scale (CD-RISC, Connor & Davidson, 2003). Genet and Siemer (2011) found that higher trait resilience was associated with better cognitive flexibility, as measured using a task-switching task. In this task, participants had to classify stimuli based on one of two task rules. Typically, a switch to a different task rule incurs a reaction time cost (the "switch cost"), and thus a greater switch cost is indicative of poorer

cognitive flexibility (Miyake et al., 2000). Similar effects were obtained by Hildebrandt et al. (2016) who showed that better resilience as indexed using the ER89 was associated with a lower switch cost (better cognitive flexibility). Likewise, Otero et al. (2020) showed that scores on the CD-RISC were associated with better scores on a neuropsychological test of cognitive flexibility using changing rule sets.

Notwithstanding these promising findings, caution is needed in interpreting these effects. Each of the above-mentioned studies used trait measures to index variation in resilience. These measures not only lack validation (Windle et al., 2011), but critically they do not conceptualize resilience as an outcome in response to adversity. Rather, the items describe personality traits thought to be characteristic of more resilient people. For example, items can describe traits such as persistence or tolerance of negative affect. This is inconsistent with contemporary conceptualizations of resilience, which highlight that resilience critically needs to be assessed by examining outcomes relative to experiences of adversity (Kalisch et al., 2019; Stainton et al., 2019).

To overcome these issues, the aim of our current study was to test the hypothesized relationship between cognitive flexibility and resilience, using a residual-based measure of resilience (Booth et al., 2022). This approach takes a measure of current emotional functioning and regresses it onto a measure of how much adversity someone has experienced. The residuals obtained from this regression constitute, for each individual, the difference between their current reported emotional functioning and the emotional functioning predicted by the adversity they have experienced. In other words, the residual-based measure of resilience reflects variation in emotional outcomes relative to what would be expected (in that sample) based on the adversity they experienced (Luthar et al., 2006). As such, this measurement approach aligns more closely with the conceptualization of resilience as an outcome relative to adversity, rather than as a trait.

The residual-based measurement approach is frequently adopted in childhood/adolescence resilience studies. For example, Bowes et al. (2010) computed a measure of childhood adversity (reports of bullying victimization) and assessed emotional problems at ages 10 and 12 through parent and teacher reports. These measures of emotional problems were regressed onto the bullying victimization measure, and the residuals were saved as indicators of resilience. Studies have shown that these residual-based measures of resilience have good predictive validity (e.g., Cahill et al., 2022), and are associated with hypothesized risk and protective factors in ways that are consistent with theory and prior empirical findings (e.g., Booth et al., 2022; Bowes et al., 2010; Kim-Cohen et al., 2004).

In our current study, participants completed a cognitive flexibility measure, as well as instruments to assess the adversity participants had experienced and current mental health difficulties. The cognitive flexibility measure was a number-letter task (Miyake et al., 2000; Rogers & Monsell, 1995), which required participants to switch between classifying numbers and classifying letters. Cognitive flexibility was assessed using the switch cost, which represents the difference in reaction times between task switches and task repetitions, with a higher score indicating less cognitive flexibility (Rogers & Monsell, 1995). To index adversity, participants completed a traumatic life events questionnaire (Vrana & Lauterbach, 1994). To capture the presence or absence of mental health difficulties, symptoms of depression, anxiety, and stress formed the measure of current mental health difficulties (Brown et al., 1997). Support for the hypothesis that cognitive flexibility is associated with variation in resilience would be obtained if a lower switch cost (representing higher cognitive flexibility) is associated with greater resilience as indexed by the residual obtained by regressing current mental health difficulties onto the traumatic life events questionnaire score.

Methods

Participants

Participants were recruited among Psychology undergraduate students at three large Australian Universities. Students participated in exchange for course credit. After data cleaning of the 812

participants who completed the study, the final sample consisted of 463 participants comprising 344 (74.5%) female, 111 (24.0%) male, and 6 (1.3%) non-binary individuals (2 missing data points). Mean age was 21.1, $SD = 5.5$. Most participants identified as White (51.3%) or Asian (32.5%), with smaller subsets identifying as African (1.7%), of mixed ethnicity (2.6%), or of another ethnicity (5.2%). According to a post-hoc power analysis in G*Power (Faul et al., 2007), a sample size of 463 in an omnibus regression with two predictors and an alpha level of .05 renders a power of .78 to detect a small effect (R^2 or $f^2 = .02$).

Materials

Adversity

Adversity was assessed using the Traumatic Events Questionnaire (TEQ; Vrana & Lauterbach, 1994). This instrument lists 11 categories of events that people can experience and that can cause serious physical, emotional, or psychological harm. The list includes both interpersonal events (e.g., being the victim of a violent crime) and non-interpersonal events (e.g., experiencing a natural disaster). Respondents are asked if they ever experienced such an event, providing a yes or no answer. The instrument has high predictive validity, and the test–retest reliability of the TEQ of specific events has been shown to range from $r = .72$ to $r = 1.00$ (Vrana & Lauterbach, 1994).

Mental health difficulties

The degree of mental health difficulties currently experienced was assessed using the 21-item version of the Depression Anxiety Stress Scale (DASS; Lovibond & Lovibond, 1995). The DASS was selected because of its transdiagnostic nature, assessing emotional difficulties across the dimensions of depression, anxiety, and stress. This general indicator of psychological distress is more appropriate to index resilience compared to instruments that focus on symptoms in only one dimension of emotional experience (Kalisch et al., 2014). Respondents rate the extent to which they experienced each symptom over the past month, using a 4-point scale ranging from 0 (Did not apply to me at all) to 3 (Applied to me most of the time). There are seven items for each dimension. Scores were summed to create a full-scale DASS score, with higher scores indicating higher overall psychological distress. The DASS-21 has good internal consistency across both clinical and non-clinical samples (Henry & Crawford, 2005; Lovibond & Lovibond, 1995) and good convergent validity with generalized psychological distress (Osman et al., 2012). In the current sample, Cronbach's alpha was .94.

Cognitive flexibility task

Cognitive flexibility was measured using the number-letter task described in Miyake et al. (2000). This task presents letter-number combinations (e.g., "A3"), and requires participants to switch between a letter task and a number task depending on where in a 2×2 grid the combination appeared. On letter trials, participants identified whether the letter was a consonant or a vowel. On number trials, participants identified whether the digit was odd or even. Cognitive flexibility can be indexed as the ease with which participants can switch from one task to the other, relative to repeating the same task (Miyake et al., 2000; Rogers & Monsell, 1995).

On each trial, the visual display consisted of a two-by-two grid, and the letter-number combination appeared in one of the four quadrants of the grid. The appearance of the letter-number combinations was predictably rotating clockwise through the grid. Whether the participant was required to perform the letter or number decision on any trial was dependent on the quadrant that the letter-number combination was presented in. For example, some participants were required to perform the letter task when the letter-number combination was presented in the top quadrants of the grid. In contrast, they would perform the number task when the letter-number combination was

presented in the bottom quadrants of the grid. The allocation of tasks to the grid side (top, bottom, left, right) was counterbalanced across participants. Participants were instructed to press the “E” key when the letter was a consonant or the number was even, and the “I” key when the letter was a vowel or number was odd (key allocation was also counterbalanced across participants).

Because of the predictable clockwise rotation through the grid, half the trials were “switch” trials, where participants performed a different task on the current trial compared to the previous trial. The other half of the trials were “repeat” trials, where participants performed the same task on the current trial as the previous trial. A Switch Cost Index was calculated by subtracting the reaction time on repeat trials from the reaction time on switch trials. A higher Switch Cost Index indicates more slowing on switch trials relative to repeat trials, and thus poorer cognitive flexibility.

Participants completed 128 critical test trials. The first quadrant position was pseudo-randomly determined, ensuring the second trial was a repeat trial. In total, there were 64 switch trials (of which 32 switches from letter to number task and 32 from number to letter task), and 64 repeat trials (32 letter task repeats and 32 number task repeats). Stimuli were drawn from the vowels A, E, I, and U, and consonants G, K, M, and R. Odd numbers were 3, 5, 7, or 9, while even numbers were 2, 4, 6, or 8. The presence of vowels/consonants and odd/even numbers was perfectly counterbalanced across trial conditions. The order of the letter and number within the combination (e.g., “A3” or “3A”) was also counterbalanced. There was a 150 ms inter-trial interval after correct responses, and a 1500 ms inter-trial-interval after incorrect responses, in order to reduce post-error slowing. The split-half reliability of the Switch Cost index (based on the Spearman-Brown corrected correlation between the Switch Cost index computed on every first versus second switch and repeat trial) was .87, indicating good internal consistency.

Procedure

This study was conducted online, with participants completing the study through an online testing platform (Inquisit 3.0.3.2, 2009). A screen calibration procedure ensured a consistent presentation of stimulus materials across varying computer displays. Participants completed a battery of questionnaires and cognitive measures as part of a larger project conducted under the Cognition Emotion Research Collaboration Initiative (CERCI).¹ Ethics approval was obtained from each University through which participants were recruited. Informed consent to participate in the study was obtained in online written form from all individual participants. The battery of questionnaires, including the DASS and the TEQ were completed first (in randomized order). This was followed by the cognitive tasks. The cognitive flexibility task was the second cognitive task to be delivered, after a short attentional control task. Before the critical block of the cognitive flexibility task, several practice blocks were presented. Participants first practiced 32 trials of the letter task and 32 trials of the number task. They then practiced the combined task in blocks of 16 trials with a key reminder when an error was made. This combined practice block was repeated until accuracy exceeded 85%, after which the critical test block commenced. If accuracy remained below 85% even after 10 combined practice blocks, the task was aborted. The experimental session lasted approximately one hour, after which participants were debriefed. The data that support the findings of this study are available at https://osf.io/zwd5v/?view_only=bf217ad1a1e746a39c125edf76c5ea11.

Results

Data cleaning and outlier analysis

Analyses were conducted using JAMOV 2.3.18 software (jamovi, 2022). Before computing the Switch Cost index, trials on which the reaction time was shorter than 100 ms, and trials on which an error was made were removed. In addition, trials with outlying reaction times—defined as deviating more than 2.5 absolute deviations from the median—were removed (Leys et al., 2013). As expected,

participants on average responded more slowly on switch trials ($M = 1264$ ms, $SD = 305$) than on repeat trials ($M = 862$ ms, $SD = 167$), $t(462) = 46.1$, $p < .001$, Cohen's $d = 2.14$. Repeat trial latencies were subtracted from switch trial latencies to compute the Switch Cost index.

Next, participant outliers were identified among the 812 participants who completed the study. Given the online nature of the study, participants were removed if they showed an outlying score on accuracy on the cognitive flexibility task (suggesting they were not properly engaged with the task), or on the Switch Cost index. Such participant outliers were defined as a score $1.5 \times IQR$ (interquartile range) above the 3rd quartile or $1.5 \times IQR$ below the first quartile (Tukey, 1977). Nineteen participants were removed because of outlying Switch Cost scores, and a further 61 were removed because of outlying accuracy scores. After the removal of these outliers, Switch Cost approximated normality, skewness = 0.523 and kurtosis = -0.214 . As resilience cannot be assessed in individuals who have not experienced significant adversity (Chmitorz et al., 2018; Kalisch et al., 2021), participants who indicated they had not experienced any traumatic life events were next excluded from further analyses. This led to the removal of a further 269 participants. The final sample consisted of 463 participants (812 minus 19 Switch Cost outliers minus 61 accuracy outliers minus 269 with no adversity experience).

Computing the residuals-based measure of resilience

The mean DASS score in the sample was 24.3 ($SD = 14.1$, range 0–63). Participants had experienced on average 2.65 traumatic events ($SD = 1.78$, range 1–10). The distribution of the number of traumatic events participants in the sample had experienced, as well as by how many participants each traumatic event was experienced can be found in Figure 1 and Table 1.

To compute a measure of adversity, the number of traumatic events people had experienced was weighted by normative scores of the impact of these events to account for some events generally having a bigger impact than others. The weights for each event (reproduced in Table 1) were based on “impact of event” scores provided by undergraduate psychology students who had listed that particular event as the worst one they had experienced (Vrana & Lauterbach, 1994). As such, the weighted total number of traumatic events participants had experienced served as the measure of adversity.

The residuals-based measure of resilience was computed by regressing the measure of current mental health difficulties (DASS scores) onto the measure of adversity (weighted TEQ scores). The

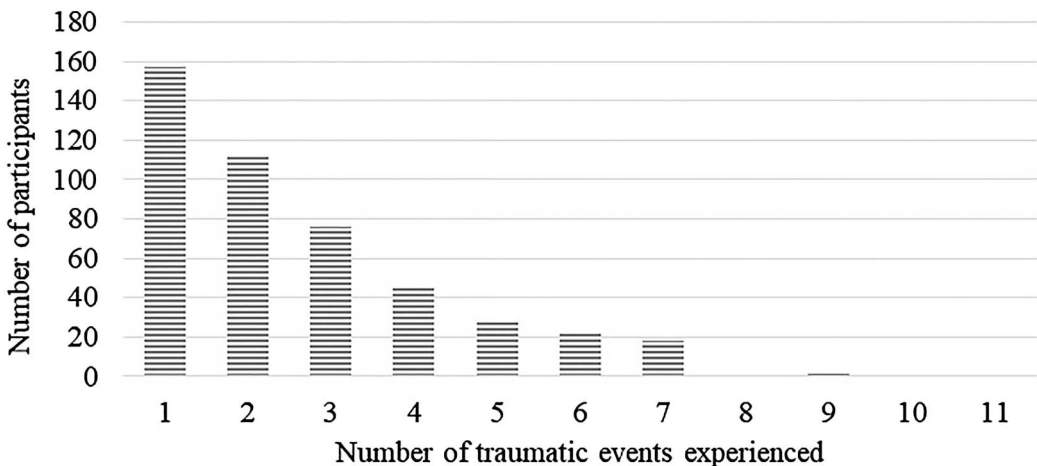


Figure 1. The distribution of the total number of traumatic events experienced by participants.

Table 1. Number and proportion of participants indicating they have experienced each item in the Traumatic Events Questionnaire, and original Impact of Event scores for each item.

Event	Number of participants indicating yes	Proportion of the sample indicating yes	Impact of Event score (Vrana & Lauterbach, 1994)
Witnessed serious accident, fire, or explosion	97	0.21	28.55
Experienced a natural disaster	61	0.13	16.3
Victim of a violent crime	82	0.18	26.1
Victim of physical or sexual abuse as a child	98	0.21	46.7
Experienced unwanted sexual experiences as an adult	88	0.19	46.1
Being in an abusive relationship as an adult	101	0.22	40.4
Witnessed mutilation, injury, or death	54	0.12	25.6
In danger of serious injury or losing life	116	0.25	26.9
Was told of mutilation, injury, or death of someone close	226	0.49	35.8
Other traumatic event like this	143	0.31	38.7
Experiences like this you can't tell about	159	0.34	43.3

model was significant, $F(1, 461) = 38.7$, $p < .001$, $R^2 = .077$, $RMSE = 13.5$, indicating that participants who experienced a greater impact of traumatic life events showed more mental health difficulties, see Figure 2.

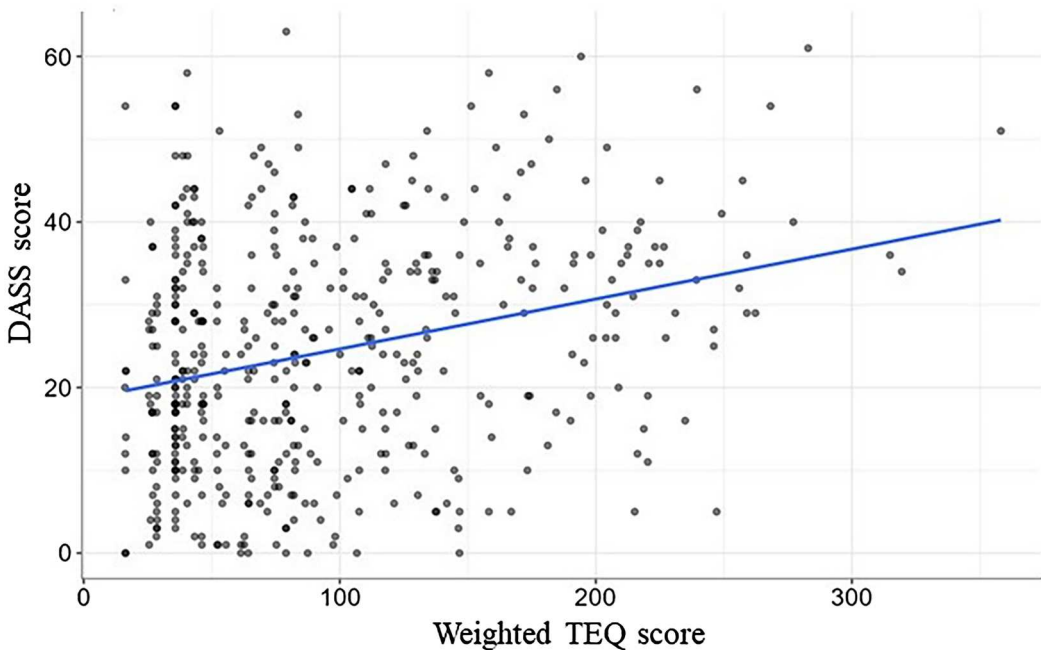
**Figure 2.** The relationship between traumatic life events experienced (weighted TEQ scores) and current mental health difficulties (DASS scores). The distance between each point and the regression line represents the residual-based resilience index. This residual was reverse scored, such that a greater score indicates fewer mental health difficulties relative to what is predicted by the adversity experienced, or greater resilience.

Table 2. Pearson correlations between main variables of interest (*p*-values between brackets).

	DASS scores	Residual-based measure of resilience	Switch Cost index
Residual-based measure of resilience	−0.961 (<i>< .001</i>)		
Switch cost index	0.082 (0.079)	−0.082 (0.078)	
Age	−0.053 (0.258)	0.092 (0.048)	−0.001 (0.986)

Note: Higher values on the switch cost index represent poorer cognitive flexibility.

The residuals of this regression, which represent the distance between an observation and the regression line, indicate the difference between observed DASS scores and DASS scores predicted by the adversity participants had experienced. These residual scores thus index the degree to which participants have more or fewer mental health difficulties relative to what is predicted by the impact of the negative life events they have experienced. A much higher DASS score than the DASS score predicted by the amount of adversity experienced indicates low resilience. A much lower DASS score than the score predicted by the adversity experienced indicates good resilience. For ease of interpretation, these residuals were reverse coded, such that higher residual scores indicate a higher level of resilience.

Is cognitive flexibility associated with resilience?

The correlations between DASS scores, the residuals-based measure of resilience, and the Switch Cost index are provided in Table 2. These correlations show that there was a small but non-significant positive association between DASS scores and the Switch Cost index and a small but non-significant negative association between the residuals-based measure of resilience and the Switch Cost index. Note that higher scores on the switch cost index represent poorer cognitive flexibility.

To test the hypothesis that better cognitive flexibility is associated with higher resilience, a linear regression analysis was conducted with the Switch Cost index as the predictor variable and the residuals-based measure of resilience as the outcome variable. We controlled for age, as there was a significant correlation between age and the residuals-based measure of resilience (see Table 2).

The overall model was significant, explaining 1.5% of the variance, $F(2, 458) = 3.50$, $p = .031$, RMSE = 13.4. After controlling for age ($\beta = .092$, $p = .047$), Switch Cost was not a significant predictor of resilience scores, $\beta = -.081$, $p = .082$. The effect size of this regression was $f^2 = .015$, which corresponds to no or a weak effect (Cohen, 1988).

The results show a high correlation between the residual-based measure of resilience and DASS scores (−.961). This high correlation is due to the statistical properties of the regression equation used to derive the residual. As this means it is difficult to differentiate between resilience and emotional vulnerability, Elman et al. (2022) suggested an alternative way to examine the role of theorized contributing factors to variation in resilience is through statistical moderation. Specifically, if a variable weakens the association between adversity and negative emotional outcomes, it can be interpreted to be associated with higher resilience. As such, we conducted a regression-based moderation analysis with weighted TEQ scores and the predictor, DASS scores as the outcome variable, and the Switch Cost index as the moderator. Results showed that the Switch Cost index did not moderate the relationship between weighted TEQ scores and DASS scores, $Z = -1.27$, $p = .204$.

Discussion

The aim of the current study was to test the hypothesis that cognitive flexibility is associated with resilience, using a residual-based measure of resilience rather than a trait measure. Our results provided no support for this hypothesis, as a Switch Cost index derived from a task-switching

paradigm was not associated with experiencing fewer mental health difficulties than what would be expected based on the traumatic events that participants had experienced. In addition, this Switch Cost index did not moderate the relationship between adversity exposure and emotional difficulties, in contrast to the notion that resilience mechanisms should weaken this relationship (Elman et al., 2022).

While our results are consistent with previous research showing no association between laboratory-based measures of cognitive flexibility and emotional vulnerability (Booth, 2014; De Lissnyder et al., 2012; Fukuzaki & Takeda, 2022; Hoffmann et al., 2017; Ruthmann et al., 2023), they are not consistent with previous research demonstrating a relationship between cognitive flexibility and resilience (Block & Kremen, 1996; Genet & Siemer, 2011; Hildebrandt et al., 2016; Otero et al., 2020). This is perhaps surprising given the magnitude of the effect size of this association observed in previous studies. For example, Otero et al. (2020) observed a partial eta squared for the difference in cognitive flexibility (measured using a task-switching paradigm) between a high and low resilient group of 0.210, which is considered a large effect (Miles & Shevlin, 2001). Genet and Siemer (2011) reported a .25 correlation between resilience and general cognitive flexibility (also measured using a task-switching paradigm), which is considered a medium-sized effect (Cohen, 1988). The absence of a significant relationship between resilience and cognitive flexibility in the current, well-powered study, stands in stark contrast to these previous findings.

One possible interpretation of this discrepancy is that conceptualizing resilience as a trait, and consequently measuring it (in part) through factors purportedly associated with this trait, leads to an over-estimation of the relationship between cognitive flexibility and resilience. For example, the CD-RISC measure of resilience includes the item “I am able to adapt when changes occur,” which could index cognitive flexibility. If so, this overlap between the outcome measure (resilience) and the predictor (cognitive flexibility), could artificially inflate the magnitude of the association. The ER89, used by Genet and Siemer (2011) in the composition of their trait resilience measure, was not developed to index whether individuals show more positive outcomes relative to what is predicted by the adversity they experienced. Instead, the ER89 was developed as a measure of ego-resiliency, which is defined as an individuals’ dynamic capacity to flexibly adapt to changes in the environment (Block & Kremen, 1996). Given this conceptual overlap between ego resiliency and cognitive flexibility, it is perhaps no surprise that a sizable association between cognitive flexibility and trait resilience was observed. In our current study, resilience was inferred from the difference between reported mental health difficulties and the mental health difficulties predicted by the adversity someone had experienced. This measurement approach thus circumvents the methodological problem of conceptual overlap between predictor and outcome which arises when using certain trait-based measures of resilience (Stainton et al., 2019).

An alternative explanation may be that different facets of cognitive flexibility are related to different manifestations of resilience. While in previous studies trait resilience was shown to be associated with cognitive flexibility measured through task-switching paradigms, it is possible that resilience as measured in the current study does not have the same underpinning mechanisms. This idea is consistent with evidence showing that executive functions such as cognitive flexibility as measured through task-switching tasks seem relatively robust against chronic stress exposure (Weckesser et al., 2021). In addition, other studies have attempted to decompose cognitive flexibility into separate constructs. For example, Kraft et al. (2020) directly compared cognitive flexibility (cue-instructed switching between two affectively neutral tasks), affective flexibility (switching between a neutral and an affective task using emotional stimuli), and feedback-based flexibility (non-cued, feedback-dependent switching between two neutral tasks). Future research should determine whether these sub-types of cognitive flexibility are related to variation in resilience.

Resilience was operationalized in the current study based on the residual-based way of measuring resilience, which has been validated in previous studies (Cahill et al., 2022) and is conceptually more robust than other existing resilience instruments especially when used in longitudinal investigations of adjustment to adversity (Kalisch et al., 2021). Despite this, future research could make changes to

the variables used in the current study to compute the residual-based resilience index, to further strengthen this measure of resilience.

Firstly, in the current study adversity was measured using a traumatic events questionnaire that queried the experience of major traumatic events across the lifetime. While the focus on major traumatic events ensures that each event qualifies as an adverse event, it has been proposed that exposure to more minor adverse events should also be taken into account as these can likewise affect mental well-being (Chmitorz et al., 2020; Kalisch et al., 2021). In addition, in the current study, we did not assess when respondents experienced each event (note the younger student sample naturally presents an upper limit to the elapsed time). Nevertheless, in an older or more varied sample, or in instances where minor events are included in an adversity checklist, it would be worthwhile assessing when each event was experienced, as literature shows that the recency of traumatic events can influence mental health outcomes (Ganzel et al., 2007; Karatzias et al., 2019). Conversely, childhood exposure to traumatic events has also been shown to have a considerable impact on mental health (Wolff & Shi, 2012). Overall, knowledge of when traumatic and other negative life events were experienced would allow future researchers to control for the age at which these events were experienced.

Secondly, in the current study, we used a measure of psychological distress to assess mental health difficulties. Other previous studies have similarly focused on the absence of mental health difficulties when capturing emotional outcomes in response to adversity (e.g., Bowes et al., 2010). However, research has shown that positive and negative affectivity are distinct dimensions of emotional experience (Diener & Emmons, 1984; Egloff, 1998; Sortheix & Weber, 2023). Moreover, the field of positive psychology assigns a central role to positive emotions in overcoming adversity and resilience (Fredrickson, 2001; Gloria & Steinhart, 2016; Ong et al., 2010; Tugade & Fredrickson, 2004). As such, future studies may wish to include measures of people's disposition to experience both positive and negative emotions to more fully capture individuals' emotional functioning relative to adversity.

While future research is needed to corroborate these findings, our results do not currently support theoretical models implicating cognitive flexibility in enhanced resilience (Kashdan & Rottenberg, 2010; Koole et al., 2015; Parsons et al., 2016; Schwager & Rothermund, 2014). Future research may seek to extend or refine the tests of these models, by incorporating recent advances in our understanding of cognitive flexibility. One avenue for future testing is based on the recognition that executive processes can be organized into hot and cold processes (Salehinejad et al., 2021; Zimmerman et al., 2016). Cold executive functioning refers to executive functioning in non-emotional situations or regarding non-emotional material (such as switching between two everyday tasks at work). Hot executive functioning on the other hand refers to executive functioning in emotionally arousing situations or regarding emotionally arousing materials (such as switching between two tasks while preparing for a stressful job interview).

Researchers have suggested that while these two types of processing are functionally interconnected, they are not entirely overlapping (Nejati et al., 2018). Focusing on cognitive flexibility more specifically, switching in the context of emotional materials is more difficult than switching in the context of non-emotional materials, as the former typically elicits a higher switch cost than the latter (D'Aurizio et al., 2022). Interestingly, there is some evidence to suggest that this increased switch cost for (negative) emotional relative to non-emotional material is especially pronounced in those with higher levels of emotional vulnerability (Deveney & Deldin, 2006). As resilience is a process that involves exposure to profound negative situations, it may be the case that an enhanced capacity for cognitive flexibility in such situations specifically contributes to better-than-expected emotional outcomes. In the current study, we only assessed cognitive flexibility in "cold" conditions, as the cognitive flexibility framework by Parsons et al. (2016) does not limit the role of cognitive flexibility in resilience to situations or stimuli that are emotionally charged. Note however that the models by Kashdan and Rottenberg (2010) and Schwager and Rothermund (2014) do typically consider flexibility in more emotional contexts (i.e., "hot" conditions). A recent study has examined the

relationship between hot and cold cognitive flexibility and trait resilience, and found that only hot cognitive flexibility (indexed through a switch cost) explained unique variance in trait resilience, but not in resilience conceptualized as an outcome (Rademacher et al., 2023). In this study, hot cognitive flexibility was assessed by requiring participants to switch between a neutral gender task (is the face depicted male or female) and an affective task (does the face show a positive or negative emotion). The task of assessing cold cognitive flexibility required switching between judgements of digit parity and magnitude. The differing tasks across these conditions make it difficult to interpret the relationship between the switch costs derived from these conditions and resilience. It is possible that the unique contribution of the hot switch cost to resilience is due to the involvement of an affective task; however, it could also be due to the difference between the two tasks in each condition. For example, it is possible there is a greater discrepancy between the difficulty of the gender task and emotion task than between the difficulty of the parity and magnitude task. To rule out this confound, this discrepancy must be matched between the two conditions, for example by holding the tasks consistent and only varying the nature of the stimuli depicted. As such, future research should therefore continue to investigate whether hot rather than cold cognitive flexibility may make a significant contribution to explaining individual differences in resilience.

Limitations and future research directions

Despite its strengths, this study is not without its limitations. One limitation concerns our restricted sample of undergraduate students, which limits the generalisability of findings. Despite the inevitable homogeneity associated with selecting from an undergraduate sample, our study sample recruited across three universities did show variation in age and ethnicity. In addition, undergraduate samples are particularly suitable for the investigation of resilience because students attending university are experiencing a critical transition period that can generate many positive but also stressful developmental challenges (Arnett, 2003). Indeed, emerging adulthood is a critical phase for establishing future trajectories in terms of educational, occupational and social attainment (Schulenberg et al., 2004). Combined with the large heterogeneity in the experiences of this age group (Arnett, 2003) as also observed in the variation in TEQ scores in our data, this developmental stage presents a pertinent backdrop for the investigation of resilience. Nevertheless, it must be noted that the range in cognitive performance in university students will be more restricted than the range which can be observed in community samples with a more varied age profile and educational background. It is possible that sample differences in variability in cognitive performance may in part explain differences in associations between resilience and cognitive flexibility observed across different studies.

An additional limitation lies in the online nature of our study, which afforded less control over the settings and circumstances in which participants completed the tasks. While participants who did not comply with task instructions in a way that affected performance on the task were identified and removed in the data cleaning process, it is still possible that other external factors may have had a more subtle influence on their performance. As such, future research may seek to replicate this study in a laboratory setting.

A third limitation concerns our single-time point assessment of resilience. Resilience is considered a dynamic concept that can change in time and with experiences. Certain experiences can represent “turning points,” providing individuals with new tools and opportunities to achieve better-than-expected outcomes (Rutter, 2012). In the case of ongoing adversity, resilience can also improve or deteriorate over time (Popham et al., 2022). This cross-sectional study is unable to capture the dynamic nature of resilience. In addition, when examining the relationship between cognitive flexibility and resilience, cross-sectional studies cannot make inferences about the causal nature of this relationship. While cognitive models of resilience generally propose that increased cognitive flexibility can lead to more resilience (e.g., Parsons et al., 2016), it is possible that the increased burden of emotional distress in less resilient individuals causes poorer cognitive flexibility

(Eysenck et al., 2007; Van Bockstaele et al., 2014). As such, future research should consider employing experimental or longitudinal designs to examine whether changes in cognitive flexibility align with changes in individuals' resilience, and/or vice versa (Kalisch et al., 2021). In addition, future research may wish to control for the effects of state mood on the relationship between resilience and cognitive flexibility.

Conclusion

In conclusion, our study in a student sample did not provide support for an association between cognitive flexibility as demonstrated in a non-emotional behavioral task, and variation in resilience conceptualized as an outcome rather than a trait. Future research could seek to refine the residual-based measure of resilience, as well as investigate the contribution of cognitive flexibility in emotionally charged situations or when processing emotional material. Ultimately, this research can assist in generating the knowledge required to increase resilience in individuals facing adversity.

Note

1. The complete battery included measures of cognitive flexibility, resilience, attentional control, attentional bias, interpretation bias, emotion regulation, worry, life satisfaction, and sleep.

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